

E-COMMERCE SALES PERFORMANCE ANALYSIS

A Business Intelligence Project using Python, SQL & Power BI

Analysis Period: January 2026 – June 2026

Project Report

Prepared for Business Performance Review

1. Dataset Description

This dataset contains 1,000 simulated e-commerce sales transactions recorded between January 2026 and June 2026. It was designed as a practice dataset for data analysis, business intelligence, and data visualization work using Python, SQL, and Power BI.

Each transaction record includes the following fields:

Field	Description
Order_ID	Unique identifier for each order
Product	Name of the purchased product
Category	Product category
Quantity	Number of units purchased
Price	Price per unit
City	Customer's city
Date	Order date

Using this dataset, the following categories of analysis were performed:

- Revenue Analysis
- Monthly Sales Trend Analysis
- Category-wise Revenue Analysis
- City-wise Sales Analysis
- Top-Selling Products
- Quantity Sold Analysis
- KPI Generation
- Data Visualization with Python and Power BI

2. Business Problem

Problem Statement

The e-commerce company experienced rapid growth in online sales during the first half of 2026 but lacked a centralized reporting system to monitor business performance. Sales data was scattered across multiple transactions, making it difficult for management to answer key business questions, including

which categories and products generate the most revenue, which cities drive growth, and how sales trends shift month to month.

Without a clear analytical dashboard, decision-makers struggled to identify high-performing products, profitable markets, and seasonal sales patterns — limiting their ability to make data-driven decisions.

Project Objective

The objective of this project is to analyze e-commerce sales data from January 2026 to June 2026 and transform raw transaction data into meaningful business insights using Python, SQL, and Power BI. The analysis focuses on identifying revenue trends, top-performing products, category performance, regional sales distribution, and key business KPIs through interactive dashboards and visualizations.

3. Methodology

The analysis followed a structured data-to-decision workflow across three tools:

- **Python:** Used for data cleaning, exploratory data analysis (EDA), correlation analysis, and static chart generation (revenue by city, revenue by category, top products, monthly trend, and a quantity–price–revenue correlation matrix).
- **SQL:** Used to aggregate transaction-level data and calculate advanced metrics — including month-over-month revenue growth and cumulative revenue — using window functions (LAG, SUM OVER).
- **Power BI:** Used to build two interactive dashboards — an Executive Sales Overview and a Product & Regional Performance Analysis — with slicers for Month, City, and Category, enabling dynamic drill-down.

4. Key Business Questions

Q: What is the total revenue generated during the analysis period?

A: ₹27.62M was generated in total revenue between January and June 2026.

Q: How many orders were placed?

A: A total of 1,000 orders were placed during the six-month period.

Q: Which product categories generated the highest revenue?

A: Electronics generated the highest revenue by a wide margin — ₹21.6M, or 78.2% of total revenue — followed by Home Appliances (11.4%), Accessories (5.9%), Fashion (4.2%), and Books (0.3%).

Q: Which are the top five revenue-generating products?

A: Laptop (₹11.41M), Tablet (₹5.93M), Smartphone (₹3.54M), Air Fryer (₹1.53M), and Watch (₹1.20M) — together accounting for roughly 88% of total revenue.

Q: Which cities contributed the most to overall sales?

A: Pune (₹3.26M), Mumbai (₹3.21M), Jaipur (₹3.16M), and Bangalore (₹3.15M) led all cities, with Hyderabad (₹1.9M) and Ahmedabad (₹2.1M) trailing furthest behind.

Q: How did monthly revenue change between January and June 2026?

A: Revenue peaked in January (₹5.21M), dipped in February (₹4.42M), recovered through March–April (~₹4.67M–4.66M), rose again in May (₹4.86M), then fell sharply in June to ₹3.81M — the lowest point in the period.

Q: What is the total quantity of products sold?

A: 3,035 units were sold in total, led by Electronics (1,017 units) and Fashion (679 units).

Q: Which categories require improvement based on revenue performance?

A: Books (0.3% of revenue) and Fashion (4.2% of revenue despite high order volume) show the weakest revenue efficiency and are candidates for strategic review.

5. Dashboard Overview

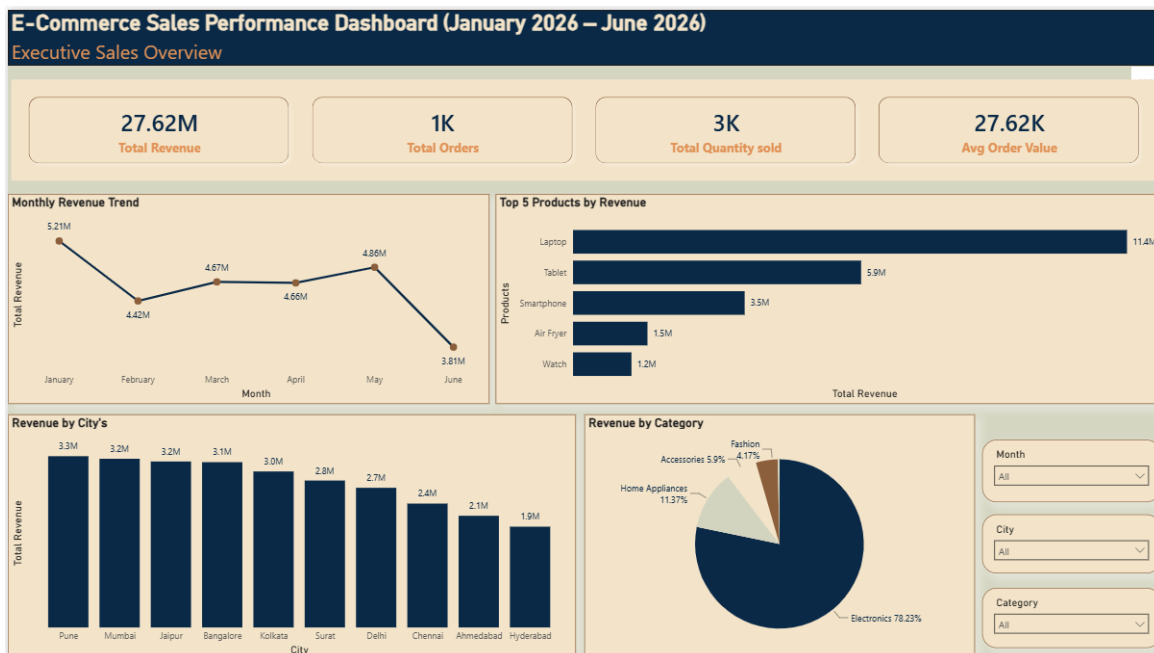
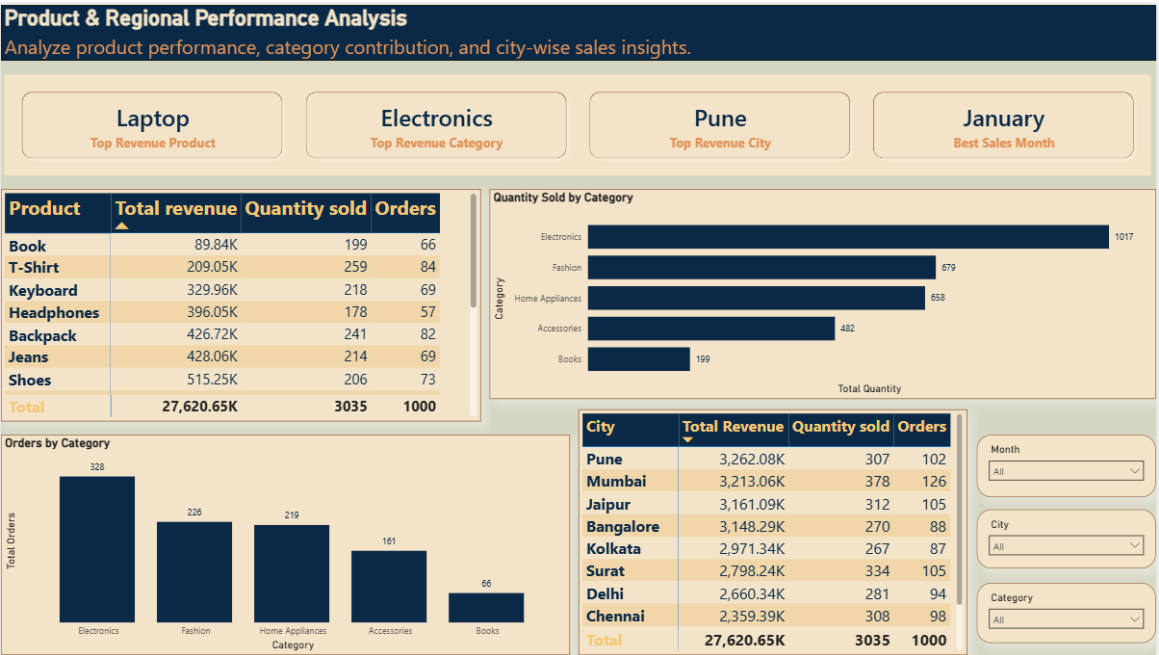


Figure 1: Executive Sales Overview Dashboard — total revenue, orders, monthly trend, top products, revenue by city, and revenue by category.



6. SQL Analysis

Beyond basic aggregation, SQL window functions were used to answer two advanced business questions: how revenue changed from one month to the next, and how revenue accumulated over the analysis period. Both queries were run against the sales table.

6.1 Month-over-Month Revenue Growth

This query calculates the percentage change in revenue between each month and the month before it, using a CTE and the LAG() window function.

Query:

```
-- Calculate month-over-month revenue growth.
WITH curr_month AS (
    SELECT month, SUM(Total_price) AS curr_month_revenue
    FROM sales
    GROUP BY month
),
prev_month AS (
    SELECT month, curr_month_revenue,
           LAG(curr_month_revenue, 1) OVER (ORDER BY month ASC) AS prev_month_revenue
    FROM curr_month
)
SELECT month, curr_month_revenue, prev_month_revenue,
       ROUND(((curr_month_revenue - prev_month_revenue) / prev_month_revenue) * 100, 2) AS
mom_growth
FROM prev_month
ORDER BY mom_growth ASC;
```

Output:

Month	Current Month Revenue	Previous Month Revenue	MoM Growth (%)
April	4,655,425	—	—
June	3,812,836	5,205,217	-26.75
February	4,418,597	4,655,425	-5.09
May	4,860,088	4,668,487	4.10
January	5,205,217	4,418,597	17.80
March	4,668,487	3,812,836	22.44

Note: Because the month field is stored as text, LAG() ordered rows alphabetically rather than chronologically, so the output above reflects alphabetical, not calendar, sequence. Re-mapped to actual calendar order, the corrected month-over-month growth is:

Month	Revenue	MoM Growth (%)
January	5,205,217	—

February	4,418,597	-15.11
March	4,668,487	5.65
April	4,655,425	-0.28
May	4,860,088	4.40
June	3,812,836	-21.55

The corrected view confirms the sharpest movements were a 15.1% drop in February and a 21.6% drop in June — the two months requiring the closest business review.

6.2 Cumulative Monthly Revenue

This query tracks running total revenue across the analysis period using SUM() with an OVER clause.

Query:

```
WITH monthly_revenue AS (
  SELECT month, SUM(Total_price) AS total_revenue
  FROM sales
  GROUP BY month
)
SELECT month, total_revenue,
       SUM(total_revenue) OVER (ORDER BY month ASC) AS cumulative_monthly_revenue
FROM monthly_revenue;
```

Output:

Month	Total Revenue	Cumulative Revenue
April	4,655,425	4,655,425
February	4,418,597	9,074,022
January	5,205,217	14,279,239
June	3,812,836	18,092,075
March	4,668,487	22,760,562
May	4,860,088	27,620,650

Note: As with the previous query, the running total accumulates in alphabetical month order rather than calendar order. The final cumulative figure (₹27,620,650) correctly matches total revenue for the period; the corrected calendar-order progression is:

Month	Total Revenue	Cumulative Revenue
January	5,205,217	5,205,217
February	4,418,597	9,623,814
March	4,668,487	14,292,301

April	4,655,425	18,947,726
May	4,860,088	23,807,814
June	3,812,836	27,620,650

This confirms revenue accumulation was steady through May, before the June shortfall reduced overall period growth.

6.3 Top 5 Revenue-Generating Products

This query identifies the five products contributing the most total revenue across the entire dataset.

Query:

```
-- Which are the Top 5 revenue-generating products?
SELECT product, category, SUM(Total_price) AS Revenue_generated
FROM sales
GROUP BY product, category
ORDER BY Revenue_generated DESC
LIMIT 5;
```

Output:

Product	Category	Revenue Generated
Laptop	Electronics	11,411,335
Tablet	Electronics	5,933,914
Smartphone	Electronics	3,535,265
Air Fryer	Home Appliances	1,533,421
Watch	Accessories	1,203,726

All top three products belong to Electronics, reinforcing the earlier finding that a small set of high-ticket items — rather than category breadth — drives the majority of revenue.

6.4 Category Contribution to Total Revenue

This query calculates each category's revenue and its percentage share of total company revenue.

Query:

```
SELECT category, SUM(Total_price) AS Total_revenue_category,
       ROUND((SUM(Total_price) / (SELECT SUM(Total_price) FROM sales)) * 100, 2) AS
percentage_contribution
FROM sales
GROUP BY category
ORDER BY percentage_contribution DESC;
```

Output:

Category	Total Revenue	% Contribution
Electronics	21,606,523	78.23

Home Appliances	3,141,480	11.37
Accessories	1,630,444	5.90
Fashion	1,152,363	4.17
Books	89,840	0.33

Electronics alone contributes more than three-quarters of all revenue, while the bottom two categories (Fashion and Books) combined contribute less than 4.5%.

6.5 Best-Selling Product per Category

This query uses RANK() partitioned by category to find the single top revenue-generating product within each category.

Query:

```
WITH category_product_sales AS (
    SELECT category, product, SUM(Total_price) AS total_revenue,
           RANK() OVER (PARTITION BY category ORDER BY SUM(Total_price) DESC) AS rnk
    FROM sales
    GROUP BY category, product
    ORDER BY total_revenue DESC
)
SELECT * FROM category_product_sales WHERE rnk = 1;
```

Output:

Category	Top Product	Revenue	Rank
Electronics	Laptop	11,411,335	1
Home Appliances	Air Fryer	1,533,421	1
Accessories	Watch	1,203,726	1
Fashion	Shoes	515,255	1
Books	Book	89,840	1

Each category has one clear revenue leader. Notably, the Books category's top (and effectively only meaningful) product — Book — accounts for its entire ₹89,840 revenue, confirming the category has almost no product depth.

7. Python Analysis

Python (pandas, seaborn, matplotlib) was used to load and clean the transaction data, engineer derived fields, and run a correlation analysis to test the relationship between quantity, price, and revenue.

7.1 Data Loading, Cleaning & Feature Engineering

The raw CSV was loaded and inspected for structure, data types, missing values, and duplicates. The Date field was converted to a proper datetime type, and two derived fields — Total_price and Month — were created to support the revenue and monthly-trend analysis used throughout this report.

Code:

```
# Initial inspection
print(df.head())
print(df.shape)

# Fix data type
df['Date'] = pd.to_datetime(df['Date'])

# Data quality checks
print(df.info())
print(df.describe())
df.isnull().sum()
df.duplicated().sum()

# Feature engineering
df['Total_price'] = df['Quantity'] * df['Price']
df['Month'] = df['Date'].dt.month_name()
```

This step confirmed the dataset had 1,000 clean transaction rows with no missing values or duplicates and produced the Total_price and Month fields used in all subsequent revenue and trend analysis.

7.2 Correlation Analysis

A correlation matrix was generated across Quantity, Price, and Total_price and visualized as a heatmap to test what actually drives revenue — order volume or item price.

Code:

```
sns.heatmap(df[['Quantity', 'Price', 'Total_price']].corr(), annot=True, cmap='Blues')
plt.show()
```

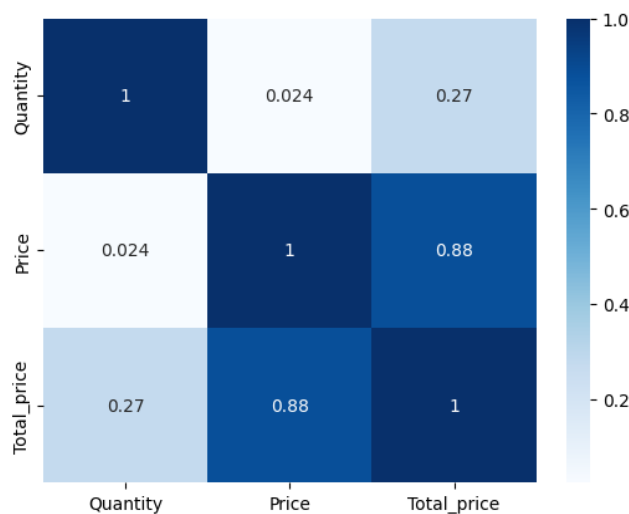


Figure 8: Correlation matrix — Quantity, Price, and Total Revenue.

Interpretation:

Variable Pair	Correlation	Meaning
Price ↔ Total_price	0.88	Strong — price is the main driver of revenue
Quantity ↔ Total_price	0.27	Weak — order volume has limited effect on revenue
Quantity ↔ Price	0.024	None — cheap and expensive items sell in similar volumes

This confirms the pattern seen throughout the dashboards and SQL analysis: revenue is price-driven, not volume-driven. High-ticket Electronics items generate the bulk of revenue despite not being purchased in disproportionately high quantities, while low-priced categories like Fashion and Accessories sell in similar or greater volume without contributing comparable revenue.

7.3 KPI Generation

Core business KPIs — total orders, category-wise product counts, total revenue, total quantity sold, and the top 10 revenue-generating products — were computed directly from the cleaned dataframe using groupby aggregations.

Code:

```
total_orders = df['Order_ID'].count()
print(f'Total Orders: {total_orders}')

total_product_per_category = df.groupby('Category')['Product'].count()
print(f'\nTotal products per Category', total_product_per_category)

Total_revenue = df['Total_price'].sum()
print(f'\nTotal revenue:', Total_revenue)

Total_quantity_sold = df['Quantity'].sum()
print(f'\nTotal quantity sold:', Total_quantity_sold)

top_products =
df.groupby('Product')['Total_price'].sum().sort_values(ascending=False).head(10)
print(f'\nTop 10 products:', top_products)
```

Output:

Metric	Value
Total Orders	1,000
Total Revenue	27,620,651.09
Total Quantity Sold	3,035

Category	Product Count
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Electronics	328
Fashion	226
Home Appliances	219
Accessories	161
Books	66

Rank	Product	Revenue
1	Laptop	11,411,335.18
2	Tablet	5,933,914.09
3	Smartphone	3,535,262.79
4	Air Fryer	1,533,421.53
5	Watch	1,203,724.83
6	Coffee Maker	812,719.87
7	Mixer	795,340.32
8	Shoes	515,253.55
9	Jeans	428,059.92
10	Backpack	426,721.67

This top-10 breakdown extends beyond the top 5 seen in the SQL analysis and dashboards, showing Coffee Maker and Mixer as the next tier of meaningful revenue contributors after the core five SKUs.

7.4 Revenue by Category (Pie Chart)

A pie chart was generated to visualize each category's share of total revenue, using the same category-revenue aggregation as the SQL contribution query in Section 6.4.

Code:

```
Revenue_by_Category = df.groupby('Category')['Total_price'].sum()
plt.pie(Revenue_by_Category.values, labels=Revenue_by_Category.index, autopct='%1.1f%%')
plt.title('Revenue By Category')
plt.show()
```

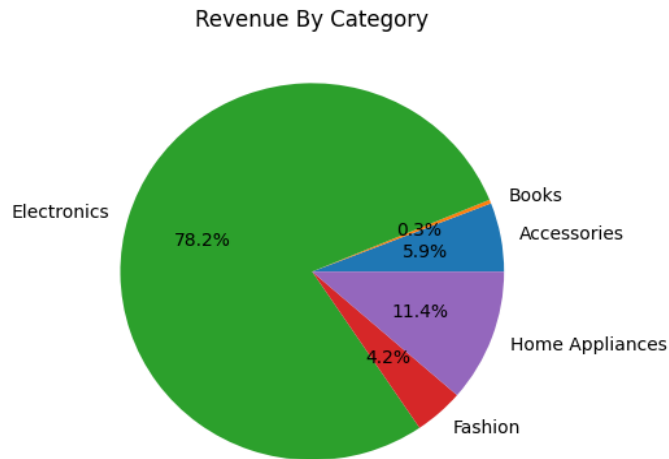


Figure 9: Revenue by Category — Electronics dominates at 78.2% of total revenue.

The chart visually reinforces the category contribution query: Electronics accounts for over three-quarters of revenue, while Books, Fashion, and Accessories together make up less than 11%.

8. Key Insights

8.1 Revenue is price-driven, not volume-driven

Correlation analysis shows that Price and Total Revenue are strongly correlated (0.88), while Quantity and Total Revenue are only weakly correlated (0.27), and Quantity and Price show almost no relationship (0.024). Revenue is being carried by high-ticket items rather than by transaction volume — Electronics makes up only 33% of orders and units but 78.2% of revenue.

8.2 A sharp revenue decline in June

The corrected SQL analysis confirms June revenue fell 21.6% versus May, and February fell 15.1% versus January — the two steepest declines in the period. June, in particular, closed the period at its lowest point and should be investigated as a priority (data completeness, stockouts, reduced marketing spend, or genuine demand drop).

8.3 High-volume categories with low revenue contribution

Fashion (226 orders, 679 units) and Accessories (161 orders, 482 units) generate strong transaction volume but contribute just 4.2% and 5.9% of revenue respectively — a classic low-average-order-value pattern.

8.4 Revenue concentration risk

The top five products (Laptop, Tablet, Smartphone, Air Fryer, Watch) generate approximately 88% of total revenue. This concentration means the business is highly exposed to supply, pricing, or demand shifts in a small number of SKUs — particularly Laptop and Tablet.

8.5 Regional performance gap

While Pune, Mumbai, Jaipur, and Bangalore perform within a tight band (₹3.1M–3.3M), Hyderabad and Ahmedabad lag 35–40% behind the top cities, pointing to an untapped growth opportunity in these markets.

9. Expected Business Outcomes

- Identification of the most profitable product categories to guide procurement and merchandising.
- Recognition of top-selling products to support inventory planning and stock prioritization.
- Determination of high-performing cities for targeted marketing and expansion campaigns.
- Ongoing monitoring of monthly sales trends, including month-over-month growth, to detect seasonal demand shifts early.
- Enablement of data-driven decision-making through interactive Power BI dashboards.
- Improved business reporting through consistent tracking of KPIs — Total Revenue, Total Orders, Quantity Sold, and Revenue by Category.

10. Recommendations

- 1. Investigate the June and February revenue declines.** Determine whether they stem from data completeness, inventory shortages, reduced marketing activity, or genuine drops in demand, and address the root cause before Q3 planning.
- 2. Reduce dependency on the top 5 SKUs.** Promote mid-tier revenue products (Coffee Maker, Mixer, Shoes, Jeans, Backpack) to build a broader revenue base and reduce concentration risk in Laptop and Tablet sales.
- 3. Use Fashion and Accessories for bundling, not standalone growth.** Pair these high-volume, low-AOV categories with high-ticket Electronics purchases to lift overall order value.
- 4. Reassess the Books category.** Decide whether it serves a strategic purpose (catalog breadth, customer acquisition) or should receive reduced inventory and marketing investment given its 0.3% revenue share.
- 5. Invest in underperforming cities.** Run a local audit of Hyderabad and Ahmedabad — delivery speed, marketing reach, and payment options — to close the 35–40% gap with top-performing cities.

6. Track revenue quality, not just volume. Since quantity sold weakly predicts revenue, prioritize KPIs like revenue-per-order and high-value SKU conversion in future reporting.

7. Fix month ordering in SQL and store dates properly. Store the date field as a proper DATE/DATETIME type (or a numeric month index) rather than a text month name, so window functions like LAG() and running SUM() order correctly without manual correction.

8. Establish a monthly trend-monitoring process. Use a rolling 3-month average with variance thresholds in the Power BI dashboard to flag sudden revenue drops earlier than they were caught this cycle.

11. Conclusion

This project successfully transformed 1,000 scattered e-commerce transactions into a structured, centralized reporting system using Python, SQL, and Power BI. The SQL window-function analysis added a time-based dimension — revealing that revenue fell 15.1% in February and 21.6% in June, with steady accumulation in between. Combined with the dashboards and correlation analysis, management now has clear visibility into revenue drivers, product and category performance, city-level distribution, and monthly trends. The most pressing findings — the June revenue shortfall and the concentration risk in a handful of top-selling products — provide immediate, actionable priorities for the second half of 2026.